JEE Main Online Exam 2022

Questions & Solution

28th July 2022 | Morning

PHYSICS

Section-A: This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.1 The dimensions of $\left(\frac{B^2}{\mu_0}\right)$ will be –

(if μ_0 : permeability of free space and B : magnetic field)

(A) $[ML^2T^{-2}]$ (B) $[MLT^{-2}]$ (C) $[ML^{-1}T^{-2}]$ (D) $[ML^2T^{-2}A^{-1}]$

Ans. **[C]**

Sol. $\left(\frac{B^2}{\mu_0}\right)$ = [Energy density]

$= \frac{ML^2T^{-2}}{L^3} = ML^{-1}T^{-2}$

Q.2 A NCC parade is going at a uniform speed of 9 km/h under a mango tree on which a monkey is sitting at a height of 19.6 m. At any particular instant, the monkey drops a mango. A cadet will receive the mango hose distance from the tree at time of drop is (Given g = 9.8 m/s^2)

(A) 5 m (B) 10 m (C) 19.8 m (D) 24.5 m

Ans. **[A]**

Sol. $H = \frac{1}{2}gt^2$

$19.6 = 4.9t^2$

$t = 2$ sec

$D = 9 \times \frac{5}{18} \times 2 = 5m$

Q.3 In two different experiments, an object of mass 5 kg moving with a speed of 25 ms^{-1} hits two different walls and comes to rest within (i) 3 second, (ii) 5 seconds, respectively.

Choose the correct option out of the following:

(A) Impulse and average force acting on the object will be same for both the cases.

(B) Impulse will be same for both the cases but the average force will be different.

(C) Average force will be same for both the cases but the impulse will be different.

(D) Average force and impulse will be different for both the cases.

Ans. **[B]**

Sol. ΔP = impulse = same since acceleration is different force acting will be different.

Q.4 A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 4.5 cm/s. If the balloon shrinks in 5 s completely. Then, the average force acting on that balloon will be (in dyne).

(A) 3 (B) 9 (C) 12 (D) 18

Ans. **[B]**

Sol. $F_{avg} = \mu \times v_{rel}$

$= \frac{10}{5} \times 4.5 = 9$

Q.5 If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately

(A) Decrease by 2% (B) Decrease by 4% (C) Increase by 2% (D) Increase by 4%

Ans. **[D]**

Sol. $g = \frac{GM}{R^2}$

$\frac{\Delta R}{R} \times 100 = -2$

$\frac{\Delta g}{g} \times \frac{2\Delta R}{R} = 4\%$

Increase by 4%

Q.6 The force required to stretch a wire of cross-section 1 cm^2 to double its length will be: (Given Young's modulus of the wire = 2×10^{11} N/m^2)

(A) 1×10^7 N (B) 1.5×10^7 N (C) 2×10^7 N (D) 2.5×10^7 N

Ans. **[C]**

Sol. $A = 1\ cm^2$

$Y = \frac{F\ell}{A\Delta\ell}$

$F = \frac{YA\Delta\ell}{\ell} = \frac{2 \times 10^{11} \times 10^{-4} \times \ell}{\ell}$

$= 2 \times 10^7$ N

Q.7 A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by 40°C, its efficiency increases by 30%. The temperature of the source will be:

(A) 166.7 K (B) 255.1 K (C) 266.7 K (D) 367.7 K

Ans. **[C]**

Sol. $1 = \frac{T_L}{T_H} = 0.5$ .... (1)

$1 - \frac{T_L - 40}{T_H} = 0.65$ .... (2)

$\Rightarrow T_H = \frac{800}{3} K \simeq 266.7K$

Q.8 Given below are two statements :

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become 2v.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

Ans. **[D]**

Sol. Average momentum = $(\vec{P}) = 0$

$v_{rms} = \sqrt{\frac{3RT}{M}}$

If temperature is doubled and oxygen atoms are used then

$V'_{rms} = \sqrt{\frac{3R(2T)}{M/2}} = 4V_{rms}$

Q.9 In the wave equation

$y = 0.5\sin\frac{2\pi}{\lambda}(400t - x)m$

the velocity of the wave will be :

(A) 200 m/s (B) $200\sqrt{2}$ m/s (C) 400 m/s (D) $400\sqrt{2}$ m/s

Ans. **[C]**

Sol. $v_{wave} = \left|\frac{\text{coefficient of t}}{\text{coefficient of x}}\right|$

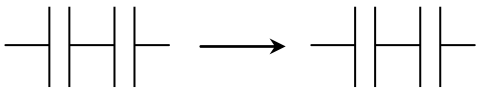
$= \frac{400}{1} = 400$ m/s

Q.10 Two capacitors, each having capacitance 40 μF are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalence capacitance of the system became 24 μF. The value of K will be :

(A) 1.5 (B) 2.5 (C) 1.2 (D) 3

Ans. **[A]**

Sol.



$\frac{40K \times 40}{40K + 40} = 24$

$40K = 24(K+1)$

$40K = 24K + 24$

$16K = 24$

$K = \frac{24}{16} = \frac{3}{2} = 1.5$

Q.11 A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is :

(A) 9 : 1 (B) 1 : 9 (C) 4 : 1 (D) 3 : 1

Ans. **[A]**

Sol. $R = \frac{\rho \ell}{A}$ ..... (i)

$A\ell$ = constant .... (ii)

$\Rightarrow R \times \ell^2$

$\Rightarrow$ Ratio = $3^2 = 9$

Q.12 The current sensitivity of a galvanometer can be increased by :
(A) Decreasing the number of turns
(B) Increasing the magnetic field
(C) Decreasing the area of the coil
(D) Decreasing the torsional constant of the spring
Choose the most appropriate answer from the options given below :
(A) (B) and (C) only (B) (C) and (D) only (C) (A) and (C) only (D) (B) and (D) only

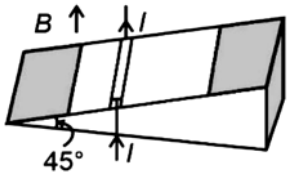
Ans. **[D]**

Sol. $NiAB = k\theta$

$\Rightarrow \frac{\theta}{i} = \frac{NiAB}{k}$

$\Rightarrow$ Sensitivity increases if B $\uparrow$ and k $\downarrow$

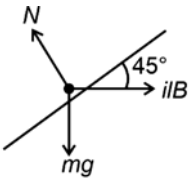
Q.13 As shown in the figure, a metallic rod of linear density 0.45 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when 0.15 T magnetic field is acting on it in the vertical upward direction, will be : {Use g = 10 m/s^2}



(A) 30A (B) 15A (C) 10 A (D) 3A

Ans. **[A]**

Sol.



$mg \times \frac{1}{\sqrt{2}} = \frac{ilB}{\sqrt{2}}$

$\Rightarrow i = \frac{mg}{B\ell}$

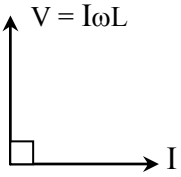
$= \frac{0.45 \times 10}{0.15} = 30A$

Q.14 The equation of current in a purely inductive circuit is $5\sin(49\ \pi t - 30°)$. If the inductance is 30 mH then the equation for the voltage across the inductor, will be : $\left\{\text{Let } \pi = \frac{22}{7}\right\}$

(A) $1.47\sin(49\ \pi t - 30°)$ (B) $1.47\sin(49\ \pi t + 60°)$
(C) $23.1\sin(49\ \pi t - 30°)$ (D) $23.1\sin(49\ \pi t + 60°)$

Ans. **[D]**

Sol.



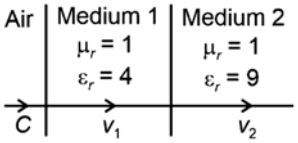
40

$V(t) = I\omega L\sin(49\pi t - 30° + 90°)$

$= 5 \times 49\pi \times \frac{30}{1000}\sin(49\pi t + 60°)$

$= 3.1\sin(49\pi t + 60°)$

Q.15 As shown in the figure, after passing through the medium 1. The speed of light v_2 in medium 2 will be : (Given $c = 3 \times 10^8$ ms^{-1})



(A) 1.0×10^8 ms^{-1} (B) 0.5×10^8 ms^{-1} (C) 1.5×10^8 ms^{-1} (D) 3.0×10^8 ms^{-1}

Ans. **[A]**

Sol. $V = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_r\varepsilon_r\mu_0\varepsilon_0}}$

$\Rightarrow v_2 = \frac{c}{\sqrt{9}} = 10^8$

Q.16 In normal adjustment, for a refracting telescope, the distance between objective and eye piece is 30 cm. The focal length of the objective, when the angular magnification of the telescope is 2, will be:

(A) 20 cm (B) 30 cm (C) 10 cm (D) 15 cm

Ans. **[A]**

Sol. $\because\ m = \frac{f_0}{f_e}$

$\Rightarrow 2 = \frac{f_0}{f_e}$ ... (i)

and, $\ell = f_0 + f_e$

$\Rightarrow 30 = f_0 + f_e$ ... (ii)

$\Rightarrow 30 = f_0 + \frac{f_0}{2}$

$\Rightarrow 30 \times \frac{2}{3} = f_0$

$\Rightarrow f_0 = 20$ cm

Q.17 The equation $\lambda = \frac{1.227}{x}\text{nm}$ can be used to find the de- Broglie wavelength of an electron. In this equation x stands for:
Where m = mass of electron
P = momentum of electron
K = Kinetic energy of electron
V = Accelerating potential in volts for electron
(A) $\sqrt{mK}$ (B) $\sqrt{P}$ (C) $\sqrt{K}$ (D) $\sqrt{V}$

Ans. **[D]**

Sol. $\because \lambda = \frac{1.227}{\sqrt{V}}\text{nm}$

$\Rightarrow x = \sqrt{V}$

Q.18 The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$th of its orignal mass to disintegrate will be -
(A) 120 days (B) 130 days (C) 180 days (D) 20 days

Ans. **[C]**

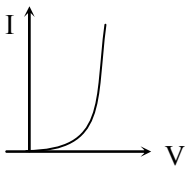
Sol. $\because N = \frac{N_0}{2^{\frac{t}{T_{1/2}}}}$

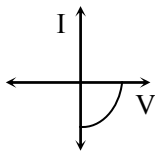
$\Rightarrow \frac{t}{2^{T_{1/2}}} = \frac{N_0}{N} = \frac{N_0}{\left(\frac{N_0}{8}\right)}$

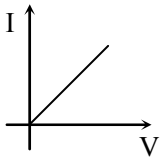
$\Rightarrow \frac{t}{2^{T_{1/2}}} = 8 = 2^3$

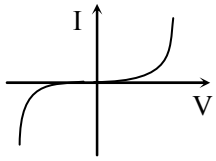
$\Rightarrow t = 3 \times T_{1/2} = 3 \times 60$
$= 180$ days

Q.19 Identify the solar cell characteristics from the following options :

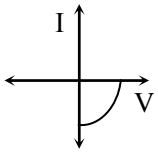
(A) 

(B) 

(C) 

(D) 

Ans. **[B]**

Sol. 

Q.20 In the case of amplitude modulation to avoid distortion the modulation index (μ) should be :
(A) $\mu \le 1$ (B) $\mu \ge 1$ (C) $\mu = 2$ (D) $\mu = 0$

Ans. **[A]**

Sol. For effective modulation,
$\mu \le 1$

Section-B: This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30). Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.21 projection of $2\hat{i}+4\hat{j}-2\hat{k}$ on $\hat{i}+2\hat{j}-\alpha\hat{k}$ is zero. Then, the value of α will be _____.

Ans. **[5]**

Sol. $\vec{A}=2\hat{i}+4\hat{j}-2\hat{k}$

$\vec{B}=\hat{i}+2\hat{j}-\alpha\hat{k}$

$\vec{A}\cdot\vec{B}=0$, as $\vec{A}$ should be perpendicular to $\vec{B}$

$\Rightarrow 2+8-2\alpha=0$

$\alpha=5$

Q.22 A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be _____ hours.

Ans. **[15]**

Sol. $T_{½}=150$ minutes

$A_0=64x$, where x is safe limit

$x=64x\times 2^{-\frac{n}{T_{½}}}$

$\Rightarrow \frac{1}{64}=2^{\frac{n}{T_{½}}}$

or $\frac{n}{T_{½}}=6$

$\Rightarrow n=6\times 150$ minutes

$=15$ hours

Q.23 In a Young's double slit experiment, a laser light of 560 nm produces an interference pattern with consecutive bright fringes' separation of 7.2 mm. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of 8.1 mm. The wavelength of second light is _____ nm.

Ans. **[630]**

Sol. $\lambda=560\times10^{-9}$

$B_1=7.2\times10^{-3}$

$B_2=8.1\times10^{-3}$

$\frac{B_1}{B_2}=\frac{\lambda_1}{\lambda_2}$

$\Rightarrow \lambda_2=\frac{560\times10^{-9}\times8.1\times10^{-3}}{7.2\times10^{-3}}$

$=6.3\times10^{-7}$

$=630$ nm

Q.24 The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are 212 rad s^{-1} and 232 rad s^{-1}. The value of resistance in the Question: circuit is R = 5 Ω. The self-inductance in the circuit is _____ mH.

Ans. **[250]**

Sol. $\frac{i}{i_{max}} = \frac{1}{\sqrt{2}}$

$= \frac{v_0/Z}{v_0/R}$

$\Rightarrow \frac{R}{Z} = \frac{1}{\sqrt{2}}$

and $\frac{1}{212C} - 212L = 232L - \frac{1}{232C}$

so $212L = \frac{1}{232C}$

so $\frac{R}{\sqrt{R^2 + \left(232L + \frac{1}{232C}\right)^2}} = \frac{1}{\sqrt{2}}$

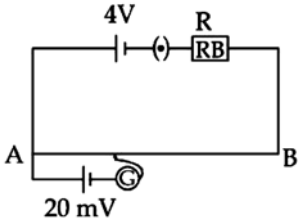
$\frac{R^2}{R^2 + (20L)^2} = \frac{1}{2}$

$L = \frac{5}{20}$

$H = \frac{5}{20} \times 1000 \text{ mH}$

$= 250 \text{ mH}$

Q.25 As shown in the figure, a potentiometer wire of resistance 20 Ω and length 300 cm is connected with resistance box (R.B.) and a standard cell of emf 4 V. For a resistance 'R' of resistance box introduced into the circuit, the null point for a cell of 20 mV is found to be 60 cm. The value of 'R ' is Ω.



Ans. **[780]**

Sol. $\ell = 3\text{m}, R_w = 20\Omega$

$\varepsilon_0 = 4\text{V}$

$\frac{4 \times 20}{20 + R} \times \frac{60}{300} = 20 \times 10^{-3}$

$\frac{4}{20 + R} = 5 \times 10^{-3}$

$20 + R = 800$

$R = 780\ \Omega$

Q.26 Two electric dipoles of dipole moments 1.2×10^{-30} C-m and 2.4×10^{-30} C-m are placed in two different uniform electric fields of strength 5×10^4 NC^{-1} and 15×10^4 NC^{-1} respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$. The value of x is _____.

Ans. **[6]**

Sol. $\frac{\tau_1}{\tau_2} = \frac{\mu_1 B_1 \sin 90}{\mu_2 B_2 \sin 90}$

$= \frac{1.2\times10^{-30}\times5\times10^{4}}{2.4\times10^{-30}\times15\times10^{4}} = \frac{1}{6}$

Q.27 The frequency of echo will be _____ Hz if the train blowing a whistle of frequency 320 Hz is moving with a velocity of 36 km/h towards a hill from which an echo is heard by the train driver. Velocity of sound in air is 330 m/s.

Ans. **[340]**

Sol. $v_s = 36\times\frac{5}{18} = 10 \text{ m/sec}$

$f = \frac{v + v_s}{v - v_s} f_0$

$= \frac{340}{320}\times320$

$= 340 \text{ Hz}$

Q.28 The diameter of an air bubble which was initially 2 mm, rises steadily through a solution of density 1750 kg m^{-3} at the rate of 0.35 cms^{-1}. The coefficient of viscosity of the solution is _____ poise (in nearest integer). (The density of air is negligible).

Ans. **[11]**

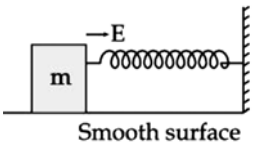
Sol. $F = 6\pi\eta rv$

$\frac{4}{3}\pi r^3\rho_1 g = 6\pi\eta rv$

$\eta = \frac{2r2\rho_1 g}{v}$

$= \frac{2\times(2\times10^{-3})^2\times1750\times10}{9\times3.5\times10^{-3}\times4} = 11 \text{ poise}$

Q.29 A block of mass 'm' (as shown in figure) moving with kinetic energy E compresses a spring through a distance 25 cm when, its speed is halved. The value of spring constant of used spring will be nE Nm^{-1} for n = _____.



Ans. **[24]**

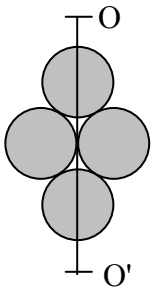
Sol. $\Delta KE = W_{all}$

So, $\frac{E}{4} - E = \frac{1}{2}K\times(0.25)^2$

$\frac{3E}{4} = \frac{1}{2}K\times\frac{1}{16}$

$k = 24\ E$

Q.30 Four identical discs each of mass 'M' and diameter 'a' are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4}ma^2$. Then, the value of x will be _____.



Ans. **[3]**

Sol. $$I = 2\times\left(\frac{M\left(\frac{a}{2}\right)^2}{4}\right)+2\times\left(\frac{M\left(\frac{a}{2}\right)^2}{4}+M\left(\frac{a}{2}\right)^2\right)$$

$$= \frac{Ma^2}{8}+\frac{5Ma^2}{8} = \frac{6Ma^2}{8} = \frac{3}{4}Ma^2$$

CHEMISTRY

Section-A: This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.1 Identify the incorrect statement from the following.
(A) A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.
(B) An orbital is the one electron wave function (Ψ) in an atom.
(C) The existence of Bohr's orbits is supported by hydrogen spectrum.
(D) Atomic orbital is characterised by the quantum numbers n and l only.

Ans. **[D]**

Sol. Atomic orbital is characterised by the quantum numbers n, l and m. Hence option D is incorrect.

Q.2 Which of the following relation is not correct?
(A) $\Delta H = \Delta U - P\Delta V$ (B) $\Delta U = q + W$ (C) $\Delta S_{sys} + \Delta S_{surr} \geq 0$ (D) $\Delta G = \Delta H - T\Delta S$

Ans. **[A]**

Sol. $\Delta H = \Delta U + P\Delta V$
Hence option A is incorrect.

Q.3 Match List-I with List-II :

	List-I		List-II
(a)	$Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$	(I)	Primary battery
(b)	$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$	(II)	Discharging of secondary battery
(c)	$2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$	(III)	Fuel cell
(d)	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	(IV)	Charging of secondary battery

Choose the correct answer from the options given below:
(A) (a)-(I), (b)-(II), (c)-(III), (d)-(IV)
(B) (a)-(IV), (b)-(I), (c)-(II), (d)-(III)
(C) (a)-(II), (b)-(I), (c)-(IV), (d)-(III)
(D) (a)-(II), (b)-(I), (c)-(III), (d)-(IV)

Ans. **[C]**

Sol. (A) $Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$
(Discharging of secondary battery)
(B) $Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$
(Primary battery)
(C) $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$
(Charging of secondary battery)
(D) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
(Fuel cell)

Q.4 Match List-I with List-II.

	List-I Reaction		**List-II** Catalyst
(a)	$4NH_3(g) + 5O_2(g) \rightarrow NO(g) + 6H_2O(g)$	(i)	$NO(g)$
(b)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(ii)	$H_2SO_4(l)$
(c)	$C_{12}H_{22}O_{11}(aq) + H_2O(l) \rightarrow \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$	(iii)	$Pt(s)$
(d)	$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	(iv)	$Fe(s)$

Choose the correct answer from the options given below:
(A) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
(B) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(C) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(D) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Ans. **[C]**

Sol. **(A)** $4NH_3(g) + 5O_2(g) \xrightarrow{Pt(s)} NO(g) + 6H_2O(g)$

(B) $N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$

(C) $C_{12}H_{22}O_{11}(aq) + H_2O(l) \xrightarrow{H_2SO_4(l)} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$

(D) $2SO_2(g) + O_2(g) \xrightarrow{No(g)} 2SO_3(g)$

Q.5 In which of the following pairs, electron gain enthalpies of constituent elements are nearly the same or identical?
(a) Rb and Cs (b) Na and K (c) Ar and Kr (d) I and At
Choose the correct answer from the options given below:
(A) (a) and (b) only (B) (b) and (c) only (C) (a) and (c) only (D) (c) and (d) only

Ans. **[C]**

Sol.

Element	Electron gain enthalpy (kJ mol^{-1})
Rb	–47
Cs	–46

Electron gain enthalpy of noble gases is almost zero.
Hence the correct option is (C).

Q.6 Which of the reaction is suitable for concentrating ore by leaching process?
(A) $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$
(B) $Fe_3O_4 + CO \rightarrow 3FeO + CO_2$
(C) $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$
(D) $Al_2O_3 + 6Mg \rightarrow 6MgO + 4Al$

Ans. **[C]**

Sol. Leaching involves the treatment of ore with a suitable reagent so as it make it soluble while impurities remain insoluble.

$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)4]$

Soluble complex

Q.7 The metal salts formed during softening of hardwater using Clark's method are :
(A) $Ca(OH)_2$ and $Mg(OH)_2$
(B) $CaCO_3$ and $Mg(OH)_2$
(C) $Ca(OH)_2$ and $MgCO_3$
(D) $CaCO_3$ and $MgCO_3$

Ans. **[B]**

Sol. In Clark's method, calculated amount of lime is added to hard water. It precipitates out calcium carbonate and magnesium hydroxide which can filtered off.

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$

$Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow -2CaCO_3\downarrow + Mg(OH)_2 + 2H_2O$

Q.8 Which of the following statement is incorrect?
(A) Low solubility of LiF in water is due to its small hydration enthalpy.
(B) KO_2 is paramagnetic.
(C) Solution of sodium in liquid ammonia is conducting in nature.
(D) Sodium metal has higher density than potassium metal.

Ans. **[A]**

Sol. Low solubility of LiF in water is due to the fact that though Li^+ is having high hydration enthalpy but it has higher lattice enthalpy when present in LiF. Due to higher lattice enthalpy its solubility is less.

Q.9 Match List-I with List-II, match the gas evolved during each reaction.

	List-I		List-II
(a)	$(nH_4)_2\ Cr_2O_2 \xrightarrow{\Delta}$	(i)	H_2
(b)	KMnO4 + HCl $\longrightarrow$	(ii)	N_2
(c)	Al + NaOH + H_2O $\longrightarrow$	(iii)	O_2
(d)	$NaNO_3 \xrightarrow{\Delta}$	(iv)	Cl_2

Choose the correct answer from the options givenbelow:
(A) (a)-(II), (b)-(III), (c)-(I), (d)-(IV)
(B) (a)-(III), (b)-(I), (c)-(IV), (d)-(II)
(C) (a)-(II), (b)-(IV), (c)-(I), (d)-(III)
(D) (a)-(III), (b)-(IV), (d)-(I), (d)-(II)

Ans. **[C]**

Sol. $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + 4H_2O + Cr_2O_3$

$KMnO_4 + HCl \longrightarrow KCl + MnCl_2 + Cl_2 + H_2O$

$Al + NaOH + H_2O \longrightarrow Na(Al(OH)_4) + H_2$

$2NaNO_3(s) \xrightarrow{\Delta} 2NaNO_2(s) + O_2$

Q.10 Which of the following has least tendency to liberate H_2 from mineral acids?
(A) Cu (B) Mn (C) Ni (D) Zn

Ans. **[A]**

Sol. The metal atom whose oxidation potential is less than that of hydrogen can release H_2 from mineral acids.

$E^\circ_{zn/zn^{+2}} = 0.76$ $E^\circ_{Ni/Ni^{+2}} = 0.25$

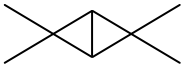
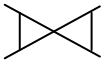
$E_{Mn/Mn^{+2}} = 1.18$ $E^\circ_{Cu/Cu^{+2}} = -0.34$

Q.11 Given below are two statements:
Statement I: In polluted water values of both dissolved oxygen and BOD are very low.
Statement II: Eutrophication results in decrease in the amount of dissolved oxygen. In the light of the above statements, choose the most appropriate answer from the options given below:
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

Ans. **[D]**

Sol. Clean water would have BOD value of less than 5 ppm whereas highly polluted water could have BOD value of 17 ppm or more.
• Eutrophication results in decrease in the amount of dissolved oxygen.

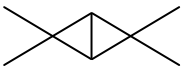
Q.12 Match the List-I with List-II .

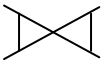
	List-I		List-II
(a)		(i)	Spiro compound
(b)		(ii)	Aromatic compound
(c)		(iii)	Non-planar Heterocyclic compound
(d)		(iv)	Bicyclo compound

(A) (a) – (II), (b) – (I), (c) – (IV), (d) – (III)
(B) (a) – (IV), (b) – (III), (c) – (I), (d) – (II)
(C) (a) – (III), (b) – (IV), (c) – (I), (d) – (II)
(D) (a) – (IV), (b) – (III), (c) – (II), (d) – (I)

Ans. **[C]**

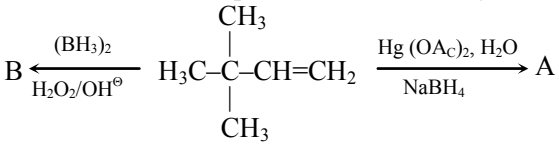
Sol. (A)  = Non-planar Heterocyclic compound

(B)  = Bicyclo compound

(C)  = Spiro compound

(D)  = Aromatic compound

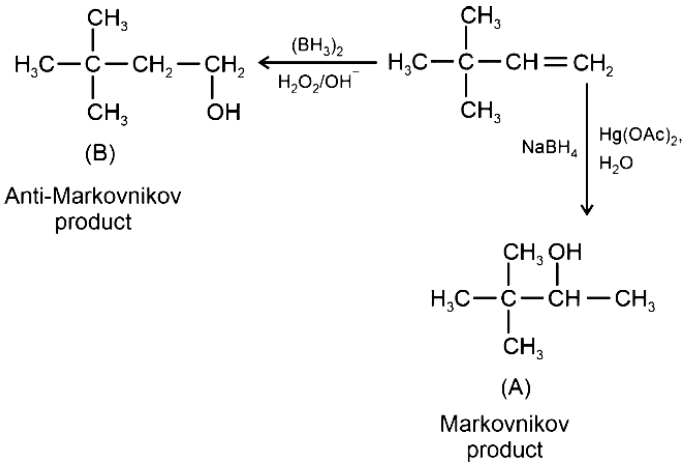
Q.13 Choose the correct option for the following reactions.



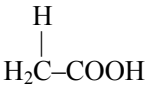
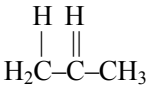
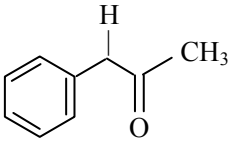
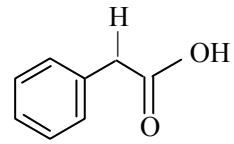
(A) 'A' and 'B' are both Markovnikov addition products
(B) 'A' is Markovnikov product and 'B' is anti- Markovnikov product
(C) 'A' and 'B' are both anti-Markovnikov products
(D) 'B' is Markovnikov and 'A' is anti-Markovnikov product

Ans. **[B]**

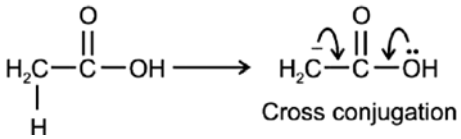
Sol.

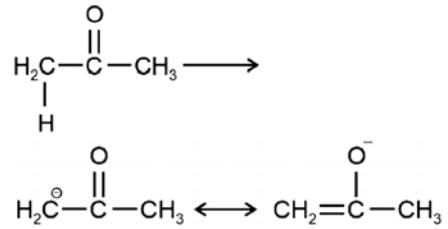


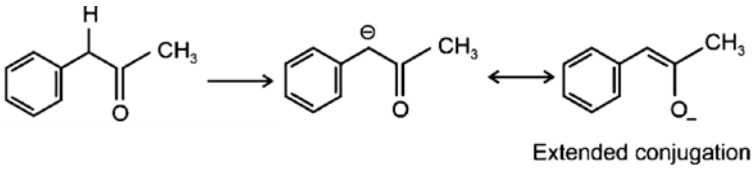
Q.14 Among the following marked proton of which compound shows lowest pk_a value ?

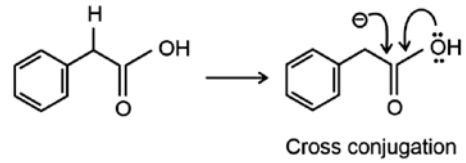
(A)  (B)  (C)  (D) 

Ans. **[C]**

Sol. (A) 

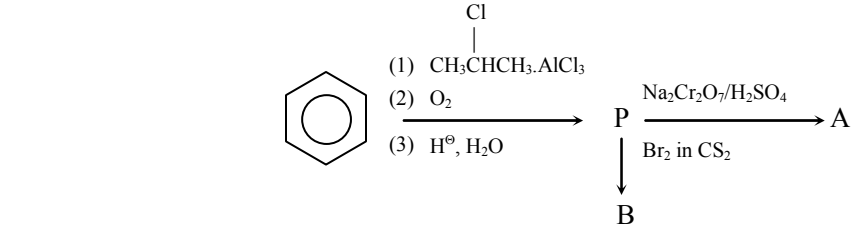
(B) 

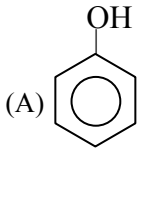
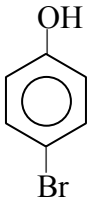
(C) 

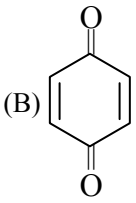
(D) 

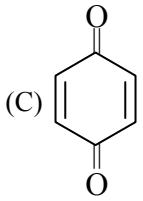
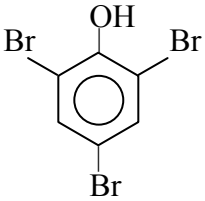
The conjugate base of compound (C) is stabilized by extended conjugation. Hence the indicated proton of compound C is most acidic i.e. will have lowest pK_a.

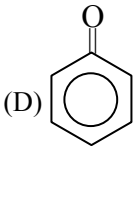
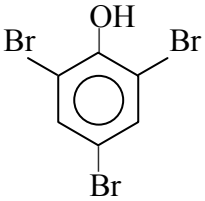
Q.15 Identify the major products A and B for the below given reaction sequence.



(A)  and 

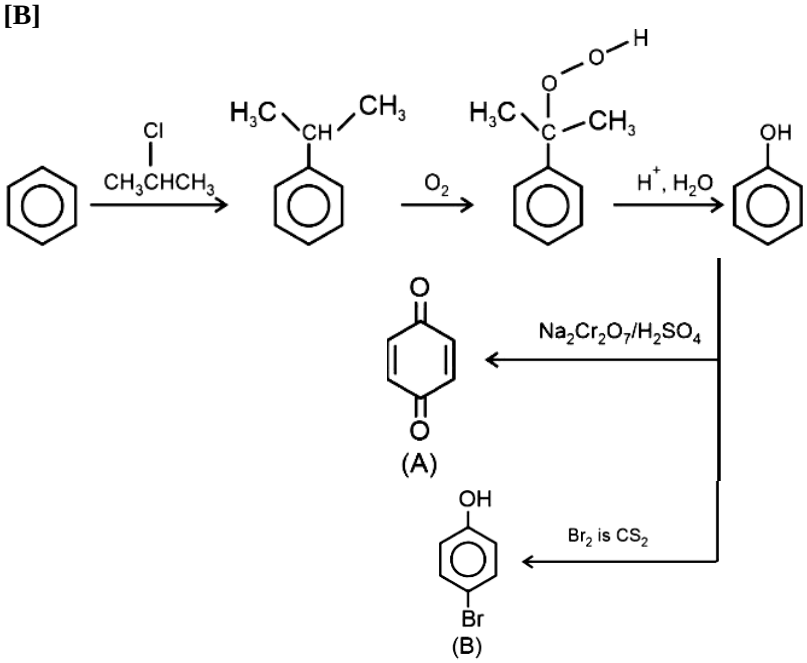
(B)  and 

(C)  and 

(D)  and 

Ans. **[B]**

Sol.



Q.16 Identify the correct statement for the below given transformation.

$$CH_3–CH_2–CH_2–\underset{\oplus N(CH_3)_3}{\underset{|}{CH}}–CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa} \underset{\text{(Major)}}{A} + \underset{\text{(Minor)}}{B}$$

(A) A–CH_3CH_2CH=CH–CH_3, B–$CH_3CH_2CH_2CH = CH_2$, Saytzeff products

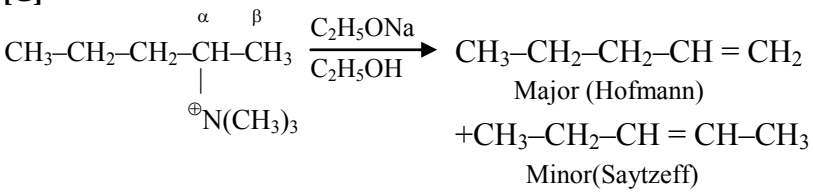
(B) A–CH_3CH_2CH=CH–CH_3, B–$CH_3CH_2CH_2CH = CH_2$, Hofmann products

(C) A–$CH_3CH_2CH_2CH$=CH_2, B–$CH_3CH_2CH = CHCH_2$, Hofmann products

(D) A–$CH_3CH_2CH_2CH$=CH_2, B-$CH_3CH_2CH = CHCH_3$, Saytzeff products

Ans. **[C]**

Sol.



Q.17 Terylene polymer is obtained by condensation of:

(A) Ethane-1, 2-diol and Benzene-1, 3 dicarboxylic acid

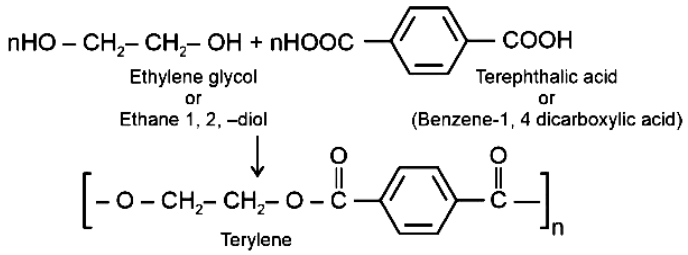
(B) Propane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

(C) Ethane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

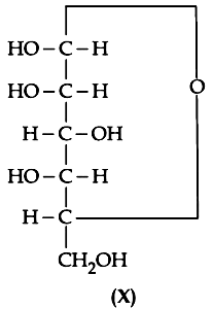
(D) Ethane-1, 2-diol and Benzene-1, 2 dicarboxylic acid

Ans. **[C]**

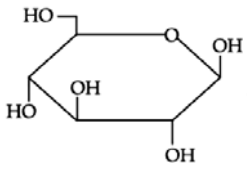
Sol.

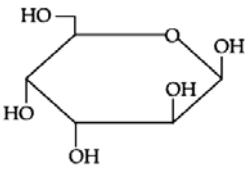


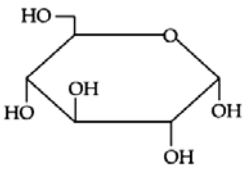
Q.18 For the below given cyclic hemiacetal (X), the correct pyranose structure is :

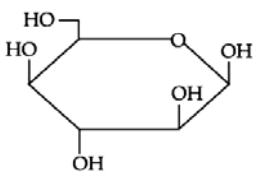


(X)

(A) 

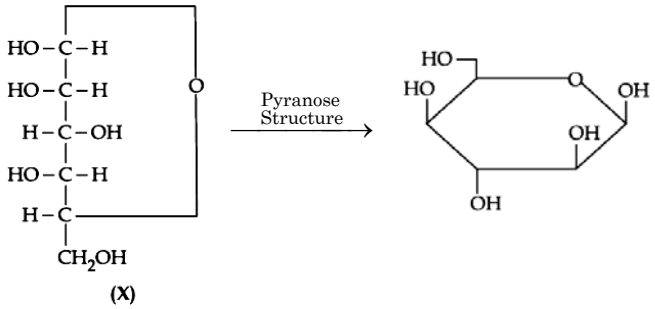
(B) 

(C) 

(D) 

Ans. **[D]**

Sol.



(X)

–OH on right side will point downwards

–OH on left side will point upwards

Q.19 Statements about Enzyme Inhibitor Drugs are given below :
(a) There are Competitive and Non-competitive inhibitor drugs.
(b) These can bind at the active sites and allosteric sites.
(c) Competitive Drugs are allosteric site blocking drugs.
(d) Non-competitive Drugs are active site blocking drugs.
Choose the correct answer from the options given below:
(A) (a), (d) only (B) (a), (c) only (C) (a), (b) only (D) (a), (b), (c) only

Ans. **[c]**

Sol. Drugs can inhibit the attachment of substrate on active site of Enzyme in two ways.
(1) Competitive, (2) Non-competitive
Competitive inhibitors bind on the active site of Enzymes. Non-Competitive inhibitors bind on allosteric site.

Q.20 For kinetic study of the reaction of iodide ion with H_2O_2 at room temperature :
(a) Always use freshly prepared starch solution.
(b) Always keep the concentration of sodium thiosulphate solution less than that of Kl solution.
(c) Record the time immediately after the appearance of blue colour.
(d) Record the time immediately before the appearance of blue colou
(e) Always keep the concentration of sodium thiosulphate solution more than that of Kl solution.
Choose the correct answer from the options given below :
(A) (a), (b), (c) only (B) (a), (d), (e) only (C) (d), (e) only (D) (a), (b), (e) only

Ans. **[A]**

Sol. Sol. To minimize contamination, use freshly prepared starch solution to determine end point. As KI is used in excess to consume all the H_2O_2 the concentration of sodium thiosulphate solution is less than KI solution. After appearance of blue colour record the time immediately.

Section-B: This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30). Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.21 In the given reaction, $X + Y + 3Z \rightleftharpoons XYZ_3$
if one mole of each of X and Y with 0.05 mol of Z gives compound XYZ_3 (Given : Atomic masses of X, Y and Z are 10, 20 and 30 amu, respectively.) the yield of XYZ_3 is _____ g. (Nearest integer)

Ans. **[2]**

Sol. $\underset{n_{moles}=1}{X} + \underset{1}{Y} + \underset{.05}{3Z} \rightleftharpoons XYZ_3$

Liminting reagent is $Z = \frac{.05}{3} = 0.16$

3 moles of $Z \rightarrow 1$ mole of XYZ_3

.05 mole of $Z \rightarrow \frac{1}{3} \times .05$ mole of XYZ_3

Wt. of $XYZ_3 = \frac{.05}{3} \times 120$

= 2 g

Q.22 An element M crystallises in a body centred cubic unit cell with a cell edge of 300 pm. The density of the element is 6.0 g cm^{-3} . The number of atoms present in 180 g of the element is _____$\times 10^{-3}$. (Nearest integer)

Ans. **[22]**

Sol. a = 300 pm
$= 300 \times 10^{-12}$m
$= 300 \times 10^{-10}$ cm
$= 3 \times 10^{-8}$ cm
$\rho = 6$ g/cm^3
wt = 180 gm
Volume = a^3
$= (3 \times 10^{-8})^3$

Volume occupied per atom = $\frac{27 \times 10^{-24}}{2}$

$\rho = 6$ wt = 180 g

Volume = $\frac{180}{6} = 30$

Number of atoms = $\frac{30 \times 2}{27 \times 10^{-24}}$

$= \frac{60}{27} \times 10^{24}$

$= 22.2 \times 10^{23}$

Q.23 The number of paramagnetic species among the following is _____ .
B_2, Li_2, C_2, C_2^-, O_2^+ and He_2^+

Ans. **[4]**

Sol. $B_2 \rightarrow 10e^-$ Paramagnetic
$Li_2 \rightarrow 6e^-$
$C_2 \rightarrow 12e^-$
$C_2^- \rightarrow 13e^-$ Paramagnetic
$O_2^+ \rightarrow 18e^-$
$O_2^+ \rightarrow 15e^-$ Paramagnetic
$He_2^+ \rightarrow 3e^-$ Paramagnetic
Species with odd number of electrons are paramagnetic except boron and oxygen.

Q.24 150 g of acetic acid was contaminated with 10.2 g ascorbic acid ($C_6H_8O_6$) to lower down its freezing point by ($x \times 10^{-1}$)°C. The value of x is _____.(Nearest integer) (Given : K_f = 3.9 K kg mol^{-1}; molar mass of ascorbic acid = 176 g mol)$^{-1}$

Ans. **[15]**

Sol. M.wt. of acetic acid = 60g
M.wt. of acetic acid = 176g
$\Delta T_f = K_f m$

$\Delta T_f = \frac{3.9 \times 10.2 \times 1000}{176 \times 150}$

$\Delta T_f = 1.506$
$= 15.06 \times 10^{-1}$
$= 15$

Q.25 K_a for butyric acid (C_3H_7COOH) is 2×10^{-5}. The pH of 0.2 M solution of butyric acid is _____$\times10^{-1}$. (Nearest integer) (Given log2 = 0.30)

Ans. **[27]**

Sol. $Ka = C\alpha^2$ $\quad C = 0.2$ M

$\alpha = \sqrt{\frac{K_a}{C}}$ $\quad Ka = 2 \times 10^{-5}$

$= \sqrt{\frac{2\times10^{-5}}{2\times10^{-1}}}$

$= 10^{-2}$

$[H^+] = C\alpha$

$= 0.2 \times 10^{-2}$

$= 2 \times 10^{-3}$

$pH = 3 - \log 2$

$= 3 - 0.30$

$= 2.7$

$pH = 24 \times 10^{-1}$

Q.26 For the given first order reaction A $\rightarrow$ B, the half life of the reaction is 0.3010 min. The ratio of the initial concentration of reactant to the concentration of reactant at time 2.0 min will be equal to _____. (Nearest integer)

Ans. **[100]**

Sol. $t_{1/2} = \frac{0.693}{K}$ $\quad t_{1/2}$ given = 0.3010

$K = \frac{0.693}{0.3010}$

$K = 2.30$

$k = \frac{2.303}{t}\log\frac{(A_0)}{(A_1)}$

$A_0 \rightarrow$ initial concentration of reactant

$A_t \rightarrow$ concentration of reactant at time t

$2.303 = \frac{2.303}{2}\log\frac{(A_0)}{(A_1)}$

$2 = \log\frac{(A_0)}{(A_1)}$

$100 = \frac{A_0}{A_1}$

Q.27 The number of interhalogens from the following having square pyramidal structure is :

ClF_3, IF_7, BrF_5, BrF_3, I_2Cl_6, IF_5, ClF, ClF_5

Ans. **[3]**

Sol. $ClF_3 \rightarrow 3\ \sigma$ bond + 2 lone pair

$IF_7 \rightarrow 7\ \sigma$ bond + 0 lone pair

$BrF_5 \rightarrow 5\ \sigma$ bond + 1 lone pair $\rightarrow$ Square pyramidal

$BrF_3 \rightarrow 3\ \sigma$ bond + 2 lone pair

$I_2Cl_6 \rightarrow 4\ \sigma$ bond + 2 lone pair

IF5 $\rightarrow 5\ \sigma$ bond + 1 lone pair $\rightarrow$ Square pyramidal

ClF $\rightarrow 1\ \sigma$ bond + 3 lone pair

$ClF_5 \rightarrow 5\ \sigma$ bond + 1 lone pair $\rightarrow$ Square pyramidal

Q.28 The disproportionation of MnO_4^{2-} in acidic medium resulted in the formation of two manganese compounds A and B. If the oxidation state of Mn in B is smaller than that of A, then the spin-only magnetic moment (μ) value of B in BM is _____. (Nearest integer)

Ans. **[4]**

Sol. $3\overset{-6}{M}nO_4^{2-} + 4H^+ \longrightarrow \overset{+4}{M}nO_2 + \overset{-7}{M}nO_4^-$

$Mn \rightarrow 4s^2 3d^5$

$Mn^{+4} \rightarrow 3d^5$

$n = 3$

$\mu = \sqrt{n(n+2)}$

$= \sqrt{3(5)}$

$= \sqrt{15}$

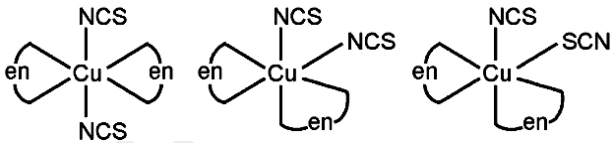
$= 3.87 \approx$ 4B.M.

Q.29 Total number of relatively more stable isomer(s) possible for octahedral complex $[Cu(en)_2(SCN)_2]$ will be _____.

Ans. **[3]**

Sol. $[Cu(en)_2(SCN)_2]$

Total isomers



Q.30 On complete combustion of 0.492 g of an organic compound containing C, H and O, 0.7938 g of CO_2 and 0.4428 g of H_2O was produced. The % composition of oxygen in the compound is _____.

Ans. **[46]**

Sol. $\%\ \text{of}\ H = \frac{2}{18} \times \frac{\text{wt. of } H_2O}{\text{wt. of organic compound}} \times 100$

$= \frac{2}{18} \times \frac{0.4428}{0.492} \times 100$

$= 0.11 \times 0.9 \times 100$

$= .099 \times 100 = 9.9$

$\%\ \text{of C} = \frac{12}{44} \times \frac{0.7938}{0.492} \times 100$

$= 0.27 \times 1.61 \times 100 = 43.47$

% Oxygen = 100 – (43.47 + 9.9)

= 100 – 53.37

= 46

MATHEMATICS

Section-A: This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.1 Let the solution curve of the differential equation $xdy = (\sqrt{x^2+y^2}+y)dx$, intersect the line x = 1 at y = 0 and the line x = 2 at y = α. Then the value of α is

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. **[B]**

Sol. $\frac{xdy-ydx}{\sqrt{x^2+y^2}} = dx$

$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{x^2+y^2}}{x} + \frac{y}{x}$

$\Rightarrow \frac{dy}{dx} = \sqrt{1+\frac{y^2}{x^2}} + \frac{y}{x}$

Let $\frac{y}{x} = v$

$\Rightarrow v + x\frac{dv}{dx} = \sqrt{1+v^2} + v$

$\Rightarrow \frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x}$

$\Rightarrow \ln(v+\sqrt{1+v^2}) = \ln x + C$

at x = 1, y = 0

$\Rightarrow C = 0$

$\frac{y}{x} + \sqrt{1+\frac{y^2}{x^2}} = x$

At x = 2,

$\frac{y}{2} + \sqrt{1+\frac{y^2}{4}} = 2$

$\Rightarrow 1+\frac{y^2}{4} = 4 + \frac{y^2}{4} - 2y$

$\Rightarrow y = \frac{3}{2}$

Q.2 Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is

(A) $\left(-\infty, \frac{1}{4}\right]$ (B) $\left[-\frac{1}{4}, \infty\right)$ (C) $\left(\frac{-1}{3}, \infty\right)$ (D) $\left(-\infty, \frac{1}{3}\right]$

Ans. **[B]**

Sol. $-1 \le \dfrac{x^2-4x+2}{x^2+3} \le 1$

$\Rightarrow -x^2-3 \le x^2-4x+2 \le x^2+3$

$\Rightarrow 2x^2-4x+5 \ge 0 \quad \& \quad -4x \le 1$

$x \in R \quad \& \quad x \ge -\dfrac{1}{4}$

So domain is $\left[-\dfrac{1}{4}, \infty\right)$

Q.3 Let the vectors $\vec{a} = (1+t)\hat{i} + (1-t)\hat{j} + \hat{k}$, $\vec{b} = (1-t)\hat{i} + (1+t)\hat{j} + 2\hat{k}$ and $\vec{c} = t\hat{i} - t\hat{j} + \hat{k}$, $t \in \mathbf{R}$ be such that for $\alpha, \beta, \gamma \in \mathbf{R}$, $\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0} \Rightarrow \alpha = \beta = \gamma = 0$. Then the set of all values of t is -

(A) A non-empty finite set (B) Equal to N

(C) Equal to $\mathbf{R} - \{0\}$ (D) Equal to $\mathbf{R}$

Ans. **[C]**

Sol. Clearly $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar

$$\begin{vmatrix} 1+1 & 1-1 & 1 \\ 1-t & 1+t & 2 \\ 1 & -t & 1 \end{vmatrix} \neq 0$$

$\Rightarrow (1+t)(1+t+2t) - (1-t)(1-t-2t) + 1(t^2-t-t-t^2) \neq 0$

$\Rightarrow (3t^2+4t+1) - (3t^2-4t+1) - 2t \neq 0$

$\Rightarrow t \neq 0$

Q.4 Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to

(A) 0 (B) 1 (C) $\dfrac{1}{2}$ (D) $-\dfrac{1}{2}$

Ans. **[A]**

Sol. $\cos^{-1}x - 2\sin^{-1}x = \cos^{-1}2x$

For Domain : $x \in \left[\dfrac{-1}{2}, \dfrac{1}{2}\right]$

$\cos^{-1}x - 2\left(\dfrac{\pi}{2} - \cos^{-1}x\right) = \cos^{-1}(2x)$

$\Rightarrow \cos^{-1}x + 2\cos^{-1}x = \pi + \cos^{-1}2x$

$\Rightarrow \cos(3\cos^{-1}x) = -\cos(\cos^{-1}2x)$

$\Rightarrow 4x^3 = x$

$\Rightarrow x = 0, \pm\dfrac{1}{2}$

Q.5 Let the operations $*, \odot \in \{\wedge, \vee\}$. If $(p * q) \odot (p \odot \sim q)$ is a tautology, then the ordered pair $(*, \odot)$ is

(A) $(\wedge, \vee)$ (B) $(\vee, \vee)$ (C) $(\wedge, \wedge)$ (D) $(\wedge, \vee)$

Ans. **[B]**

Sol. $*, \odot \in \{\wedge, \vee\}$

Now for $(p*q) \odot (p \odot \sim q)$ is tautology

(A) $(\vee, \wedge)$: $(p \vee q) \wedge (p \wedge \sim q)$ not a tautology

(B) $(\vee, \vee)$: $(p \vee q) \vee (p \vee \sim q) = p \vee T$ is tautology

(C) $(\wedge, \wedge)$: $(p \vee q) \wedge (p \wedge \sim q) = (p \wedge p) \wedge (q \wedge \sim q) = p \wedge F$ not a tautology (Fallasy)

(D) $(\wedge, \vee)$: $(p \wedge q) \vee (p \vee \sim q)$ not a tautology

Q.6 Let a vector $\vec{a}$ has magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in \mathbf{R} \times \mathbf{R} - \{(0, 0)\}$, the vector $(x\vec{a} + y\vec{b})$ is perpendicular to the vector $(6y\vec{a} - 18x\vec{b})$. Then the value of $|\vec{a} \times \vec{b}|$ is equal to -

(A) $9\sqrt{3}$ (B) $27\sqrt{3}$ (C) 9 (D) 81

Ans. **[B]**

Sol. $(x\vec{a} + y\vec{b}) \cdot (6y\vec{a} - 18x\vec{b}) = 0$

$\Rightarrow (6xy|\vec{a}|^2 - 18xy|\vec{b}|^2) + (6y^2 - 18x^2)\vec{a} \cdot \vec{b} = 0$

As given equation is identity

Coefficient of x^2 = coefficient of y^2 = coefficient of $xy = 0$

$\Rightarrow |\vec{a}|^2 = 3|\vec{b}|^2 \Rightarrow |\vec{b}| = 3\sqrt{3}$

and $\vec{a} \cdot \vec{b} = 0$

$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$

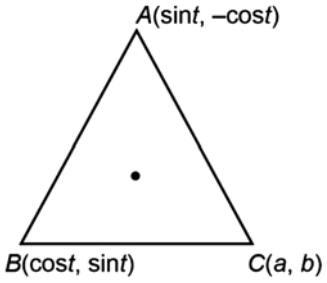
$= 9.3\sqrt{3}.1 = 27\sqrt{3}$

Q.7 For $t \in (0\ 2\pi)$, if ABC is an equilateral triangle with vertices A(sint, –cost), B (cost, sint) and C(a, b) such that its orthocentre lies on a circle with centre $\left(1, \frac{1}{3}\right)$, then $(a^2 - b^2)$ is equal to

(A) $\frac{8}{3}$ (B) 8 (C) $\frac{77}{9}$ (D) $\frac{80}{9}$

Ans. **[B]**

Sol. Let P(h, k) be the orthocentre of ΔABC



Then

$h = \frac{\sin t + \cos t + a}{3}$, $k = \frac{-\cos t + \sin t + b}{3}$

(orthocentre coincide with centroid)

$\therefore (3h-a)^2 + (3k-b)^2 = 2$

$\therefore \left(h-\frac{a}{3}\right)^2 + \left(k-\frac{b}{3}\right)^2 = \frac{2}{9}$

$\because$ orthocentre lies on circle with centre $\left(1,\frac{1}{3}\right)$

$\therefore a = 3, b = 1$

$\therefore a^2 - b^2 = 8$

Q.8 For $\alpha \in N$, consider a relation R on N given by $R = \{(x, y) : 3x + \alpha y \text{ is a multiple of } 7\}$. The relation R is an equivalence relation if and only if

(A) $\alpha = 14$ (B) α is a multiple of 4

(C) 4 is the remainder when α is divided by 10 (D) 4 is the remainder when α is divided by 7

Ans. **[D]**

Sol. $R = \{(x, y) : 3x + \alpha y \text{ is multiple of } 7\}$, Now R to be an equivalence relation

(1) R should be reflexive : $(a, a) \in R \quad a \in N$

$\therefore 3a + a\alpha = 7k$

$\therefore (3 + \alpha)\, a = 7k$

$\therefore 3 + \alpha = 7k_1 \Rightarrow \alpha = 7k_1 - 3$

$= 7k_1 + 4$

(2) R should be symmetric : $aRb \Leftrightarrow bRa$

$aRb : 3a + (7k - 3)\, b = 7\, m$

$\Rightarrow 3(a - b) + 7kb = 7\, m$

$\Rightarrow 3(b - a) + 7\, ka = 7\, m$

So, $aRb \Rightarrow bRa$

$\therefore$ R will be symmetric for $a = 7k_1 - 3$

(3) Transitive : Let $(a, b) \in R, (b, c) \in R$

$\Rightarrow 3a + (7k - 3)b = 7k_1$ and

$3b + (7k_2 - 3)\, c = 7k_3$

Adding $3a + 7kb + (7k_2 - 3)\, c = 7\,(k_1 + k_3)$

$3a + (7k_2 - 3)\, c = 7\, m$

$\therefore (a, c) \in R$

$\therefore$ R is transitive

$\therefore \alpha = 7k - 3 = 7k + 4$

Q.9 Out of 60% female and 40% male candidates appearing in an exam, 60% candidates qualify it. The number of females qualifying the exam is twice the number of males qualifying it. A candidate is randomly chosen from the qualified candidates. The probability, that the chosen candidate is a female, is

(A) $\frac{3}{4}$ (B) $\frac{11}{16}$ (C) $\frac{23}{32}$ (D) $\frac{13}{16}$

Ans. **[Bonus]**

Sol. $P(\text{Female}) = \frac{60}{100} = \frac{3}{5}$

$P(\text{Male}) = \frac{2}{5}$

$P(\text{Female / Qualified}) = \frac{40}{60} = \frac{2}{3}$

$P(\text{male / Qualified}) = \frac{20}{60} = \frac{1}{3}$

Q.10 If $y = y(x),\ x \in \left(0, \frac{\pi}{2}\right)$ be the solution curved of the differential equation $(\sin^2 2x)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y = 2e^{-4x}(2\sin 2x + \cos 2x)$, with $y\left(\frac{\pi}{4}\right) = e^{-\pi}$, then $y\left(\frac{\pi}{6}\right)$ is equal to -

(A) $\frac{2}{\sqrt{5}}e^{-2\pi/3}$ (B) $\frac{2}{\sqrt{3}}e^{2\pi/3}$ (C) $\frac{1}{\sqrt{3}}e^{-2\pi/3}$ (D) $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Ans. **[A]**

Sol. $(\sin^2 2x)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y = 2e^{-4x}(2\sin 2x + \cos 2x)$

$$\frac{dy}{dx} + (8 + 4\cot 2x)y = 2e^{-4x}\left(\frac{2\sin 2x + \cos 2x}{\sin^2 2x}\right)$$

integration factor

$(I.F.) = e^{\int (8+4\cot 2x)dx}$

$= e^{8x + 2\ln \sin 2x}$

Solution of differential equation

$y \cdot e^{8x+2\ln \sin 2x}$

$= \int 2e^{(4x+2\ln \sin 2x)} \frac{(2\sin 2x + \cos 2x)}{\sin^2 2x} dx$

$= 2\int e^{4x}(2\sin 2x + \cos 2x)dx$

$y \cdot e^{8x+2\ln \sin 2x} = e^{4x}\sin 2x + c$

$y\left(\frac{\pi}{4}\right) = e^{-\pi}$

$e^{-\pi} \cdot e^{2x} = e^{\pi} + c \Rightarrow c = 0$

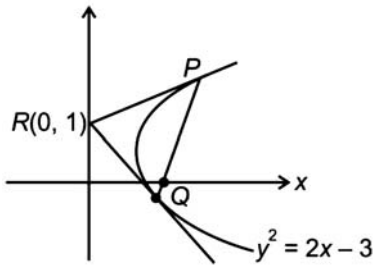
$$y\left(\frac{\pi}{6}\right) = \frac{e^{\frac{2\pi}{3}}\frac{\sqrt{3}}{2}}{e^{\left(\frac{4\pi}{3} + 2\ln\frac{\sqrt{3}}{2}\right)}} = e^{-\frac{2\pi}{3}}\frac{2}{\sqrt{3}}$$

Q.11 If the tangents drawn at the points P and Q on the parabola $y^2 = 2x - 3$ intersect at the point R(0, 1), then the orthocentre of the triangle PQR is :

(A) (0, 1) (B) (2, –1) (C) (6, 3) (D) (2, 1)

Ans. **[B]**

Sol.

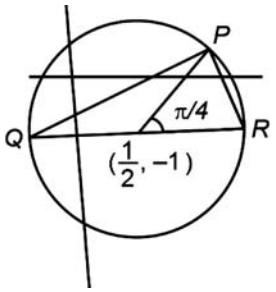


Equation of chord PQ
$\Rightarrow y \times 1 = x - 3$
$\Rightarrow x - y = 3$
For point P & Q
Intersection of PQ with parabola P : (6, 3) Q : (2, –1)
Slope of RQ = –1 & Slope of PQ = 1
Therefore $\angle PQR = 90° \Rightarrow$ Orthocentre is at Q : (2, –1)

Q.12 Let C be the centre of the circle $x^2 + y^2 - x + 2y = \frac{11}{4}$ and P be a point on the circle. A line passes through the point C, makes an angle of $\frac{\pi}{4}$ with the line CP and intersects the circle at the Q and R. Then the area of the triangle PQR (in unit^2) is -

(A) 2 (B) $2\sqrt{2}$ (C) $8\sin\left(\frac{\pi}{8}\right)$ (D) $8\cos\left(\frac{\pi}{8}\right)$

Ans. **[B]**
Sol.



QR = 2r = 4
$P = \left(\frac{1}{2} + 2\cos\frac{\pi}{4}, -1 + 2\sin\frac{\pi}{4}\right)$
$= \left(\frac{1}{2} + \sqrt{2}, -1 + \sqrt{2}\right)$
Area of $\Delta PQR = \frac{1}{2} \times 4 \times \sqrt{2}$
$= 2\sqrt{2}$ sq. units

Q.13 The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is:
(A) 0 (B) 2 (C) 3 (D) 4

Ans. **[C]**
Sol. Let $E = 7^{2022} + 3^{2022}$
$= (15 - 1)^{1011} + (10 - 1)^{1011}$
= –1 + (multiple of 15) –1 + multiple of 10
= –2 + (multiple of 5)
Hence remainder on dividing E by 5 is 3.

Q.14 Let the matrix $A=\begin{vmatrix}0&1&0\\0&0&1\\1&0&0\end{vmatrix}$ and the matrix $B_0= A^{49} + 2A^{98}$. If $B_n = \text{Adj}(B_{n-1})$ for all $n \geq 1$, then $\det(B_4)$ is equal to :

(A) 3^{28} (B) 3^{30} (C) 3^{32} (D) 3^{36}

Ans. **[C]**

Sol. $A=\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$

$$\Rightarrow A^2=\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}\times\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$$

$$\Rightarrow A^3=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}\times\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\ell$$

Now $B_0 = A^{49} + 2A^{98} = (A^3)^{16}\cdot A + 2(A^3)^{32}\cdot A^2$

$$B_0 = A + 2a^2 = \begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}+\begin{bmatrix}0&0&2\\2&0&0\\0&2&0\end{bmatrix}=\begin{bmatrix}0&1&2\\2&0&1\\1&2&0\end{bmatrix}$$

$\Rightarrow |B_0| = 9$

Since, $B_n = \text{Adj}\ |B_{n-1}| \Rightarrow |B_n| = |B_{n-1}|^2$

Hence $|B_4| = |B_3|^2 = |B_2|^4 = |B_1|^8 = |B_0|^{16}$

$= |3^2|^{16} = 3^{32}$

Q.15 Let $S_1=\left\{z_1 \in C:|z_1-3|=\frac{1}{2}\right\}$ and $S_2=\{z_2 \in C:|z_2-|z_2+1||=z_2+|z_2-1||\}$. Then, for $z_1 \in S_1$ and $z_2 \in S_2$, the least value of $|z_2 - z_1|$ is :

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. **[C]**

Sol. $\because |Z_2+|Z_2-1||^2=Z_2-|Z_2+1||^2$

$\Rightarrow (|Z_2 + |Z_2 - 1||\ (\bar{Z}_2 + |Z_2 - 1|| = (Z_2 - |Z_2 + 1|)\ (\bar{Z}_2 - |Z_2 + 1|)$

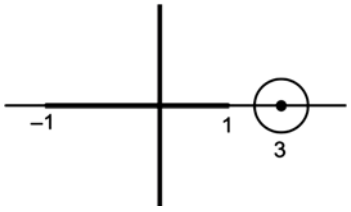
$\Rightarrow Z_2\ (|Z_2 - 1| + |z_2 + 1|) + \bar{Z}_2\ (|Z_2 - 1| + |Z_2 + 1|) = |Z_2 + 1|^2 - |Z_2 - 1|^2$

$\Rightarrow (Z_2 + \bar{Z}_2)\ (|Z_2 + 1| + |Z_2 - 1|) = 2(Z_2 + \bar{Z}_2)$

$\Rightarrow$ Either $Z_2 + \bar{Z}_2 = 0$ or $|Z_2 + 1| + |Z_2 - 1| = 2$

So, Z_2 lies on imaginary axis or on real axis with [–1, 1]

Also $|Z_1 - 3| = \frac{1}{2} \Rightarrow Z_1$ lies on the circle having center 3 and radius $\frac{1}{2}$.



Clearly $|Z_1 - Z_2|_{min} = \frac{3}{2}$

Q.16 The foot of the perpendicular from a point on the circle $x^2 + y^2 = 1$, $z = 0$ to the plane $2x + 3y + z = 6$ lies on which one of the following curves?

(A) $(6x + 5y - 12)^2 + 4(3x + 7y - 8)^2 = 1$, $z = 6 - 2x - 3y$

(B) $(5x + 6y - 12)^2 + 4(3x + 5y - 9)^2 = 1$, $z = 6 - 2x - 3y$

(C) $(6x + 5y - 14)^2 + 9(3x + 5y - 7)^2 = 1$, $z = 6 - 2x - 3y$

(D) $(5x + 6y - 14)^2 + 9(3x + 7y - 8)^2 = 1$, $z = 6 - 2x - 3y$

Ans. **[B]**

Sol. Any point on $x^2 + y^2 = 1$, $z = 0$ is p $(\cos\theta, \sin\theta, 0)$

If foot of perpendicular of p on the plane $2x + 3y + z = 6$ is (h, k, ℓ) then

$$\frac{h-\cos\theta}{2} = \frac{k-\sin\theta}{3} = \frac{\ell-0}{1} = \left(-\frac{2\cos\theta+3\sin\theta+0-6}{2^2+3^2+1^2}\right) = r(\text{let})$$

$h = 2r + \cos\theta$, $k = 3r + \sin\theta$, $\ell = r$

Hence, $h - 2\ell = \cos\theta$ and $k - 3\ell = \sin\theta$

Hence $(h - 2\ell)^2 + (k - 3\ell)^2 = 1$

When $\ell = 6 - 2h - 3k$

Hence required locus is

$(x - 2(6 - 2x - 3y))^2 + (y - 3(6 - 2x - 3y))^2 = 1$

$\Rightarrow (5x + 6y - 12)^2 + 4(3x + 5y - 9)^2 = 1$, $z = 6 - 2x - 3y$

Q.17 If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}$. $x > 0$ is 14, then the value of α is equal to -

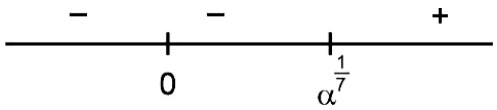
(A) 32 (B) 64 (C) 128 (D) 256

Ans. **[C]**

Sol. $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5} \{x > 0\}$

$f'(x) = 5x - \frac{5\alpha}{x^6} = 0$

$\Rightarrow x = (\alpha)^{\frac{1}{7}}$



$f(x)_{min} = \frac{5(\alpha)^{\frac{2}{7}}}{2} + \frac{\alpha}{\alpha^{\frac{5}{7}}} = 14$

$\frac{5}{2}\alpha^{\frac{2}{7}} + \alpha^{\frac{2}{7}} = 14$

$\frac{7}{2}\alpha^{\frac{2}{7}} = 14$

$\alpha = 128$

Q.18 Let α, β and γ be three positive real numbers. Let $f(x) = \alpha x^5 + \beta x^3 + \gamma x$, $x \in \mathbf{R}$ and $g : \mathbf{R} \to \mathbf{R}$ be such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. If $a_1, a_2, a_3, \ldots, a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right)$ is equal to -

(A) 0 (B) 3 (C) 9 (D) 27

Ans. **[A]**

Sol. $f\left(g\left(\frac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right)$

$\frac{a_1 + a_2 + a_3 + \ldots.. + a_n}{n} = 0$

$\therefore$ First and last term, second and second last and so on are equal in magnitude but opposite in sign.

$f(x) = \alpha x^5 + \beta x^3 + \gamma x$

$\sum_{i=1}^{n} f(a_i) = \alpha(a_1^5 + a_2^5 + a_3^5 + \ldots.. + a_n^5) + \beta(a_1^3 + a_2^3 + \ldots. + a_n^3) + \gamma(a_1 + a_2 + \ldots. + a_n)$

$= 0\alpha + 0\beta + 0\gamma$

$= 0$

$\therefore\ f\left(g\left(\frac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right) = \frac{1}{n}\sum_{i=1}^{n} f(a_i) = 0$

Q.19 Consider the sequence $a_1, a_2, a_3, \ldots$ such that $a_1 = 1$, $a_2 = 2$ and $a_{n+2} = \frac{2}{a_{n+1}} + a_n$ for $n = 1, 2, 3, \ldots..$

If $\left(\frac{a_1 + \frac{1}{a_2}}{a_3}\right) \cdot \left(\frac{a_2 + \frac{1}{a_3}}{a_4}\right) \cdot \left(\frac{a_3 + \frac{1}{a_4}}{a_5}\right) \ldots.. \left(\frac{a_{30} + \frac{1}{a_{31}}}{a_{32}}\right) = 2^{\alpha}({}^{61}C_{31})$, then α is equal to

(A) –30 (B) –31 (C) –60 (D) –61

Ans. **[C]**

Sol. $a_{n+2} = \frac{2}{a_{n+1}} + a_n$

$\Rightarrow a_n a_{n+1} + 1 = a_{n+1} a_{n+2} - 1$

$\Rightarrow a_{n+2} a_{n+1} - a_n \cdot a_{n+1} - 2$

For $n = 1$ $a_3a_2 - a_1a_2 = 2$

$n = 2$ $a_4a_3 - a_3a_2 = 2$

$n = 3$ $a_5a_4 - a_4a_3 = 2$

$\vdots$

$n = n$ $a_{n+2}a_{n+1} - a_n a_{n+1} = 2$

$a_{n+2}a_{n+1} = 2n + a_1a_2$

Now, $\frac{(a_1a_2 + 1)}{a_2a_3} \cdot \frac{(a_2a_3 + 1)}{a_3a_4} \cdot \frac{(a_3a_4 + 1)}{a_4a_5} \cdot \ldots.. \cdot \frac{(a_{30}a_{31} + 1)}{a_{31}a_{32}} = \frac{3}{4} \cdot \frac{5}{6} \cdot \frac{7}{8} \cdot \ldots.. \cdot \frac{61}{62} = 2^{-60}({}^{61}C_{31})$

Q.20 The minimum value of the twice differentiable function $f(x)=\int_0^x e^{x-1} f'(t)dt-(x^2-x+1)e^x$, $x \in R$, is

(A) $-\frac{2}{\sqrt{e}}$ (B) $-2\sqrt{e}$ (C) $-\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$

Ans. **[A]**

Sol. $f(x)=\int_0^x e^{x-t} f'(t)dt-(x^2-x+1)e^x$

$f(x)=e^x\int_0^x e^{-t} f'(t)dt-(x^2-x+1)e^x$

$e^{-x}f(x)=\int_0^x e^{-t} f'(t)dt-(x^2-x+1)$

Differentiate on both side

$e^{-x} f'(x) + (-f(x)\, e^{-x}) = e^{-x} f'(x) - 2x + 1$

$f(x) = e^x (2x - 1)$

$f'(x) = e^x (2) + e^x (2x - 1) = e^x (2x + 1)$

$x=-\frac{1}{2}$

$f''(x)=e^x(2)+(2x+1)e^x$

$= e^x(2x+3)$

For $x=-\frac{1}{2}$ $f''(x) > 0$

$\Rightarrow$ maxima

$\therefore$ Max. $= e^{-\frac{1}{2}}(-1-1)$

$\therefore -\frac{2}{\sqrt{e}}$

Section-B: This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30). Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

Q.21 Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from {A, B, C, D, E} or a number from {1, 2, 3, 4, 5} with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from {1, 2, 3, 4, 5} is $\alpha \times 5^6$, then α is equal to _____.

Ans. **[7073]**

Sol. If password is 6 character long, then

Total number of ways having atleast one number = $10^6 - 5^6$

Similarly, if 7 character long = $10^7 - 5^7$ and if 8-character long = $10^8 - 5^8$

Number of password = $(10^6 + 10^7 + 10^8) - (5^6 + 5^7 + 5^8) = 5^6 (2^6 + 5.2^7 + 25.2^8 - 1 - 5 - 25)$

$= 5^6(64 + 640 + 6400 - 31)$

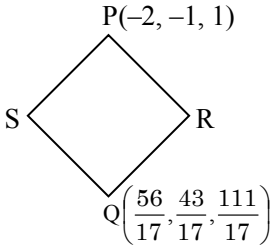
$= 7073 \times 5^6$

$\therefore \alpha = 7073$

Q.22 Let P(–2, –1, 1) and $Q\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagonal RS are α, –1, β, where both α and β are integers of minimum absolute values, then $\alpha^2 + \beta^2$ is equal to _____.

Ans. **[450]**

Sol.



d·r's of RS = $< \alpha, -1, \beta >$

d·r's of PQ = $< \frac{90}{17},\frac{60}{17},\frac{94}{17} > = < 45, 30, 47 >$

as PQ and RS are diagonals of rhombus

$\alpha(45) + 30(-1) + 47(\beta) = 0$

$\Rightarrow 45\alpha + 47\beta = 30$

i.e. $\alpha = \frac{30 - 47\beta}{45}$

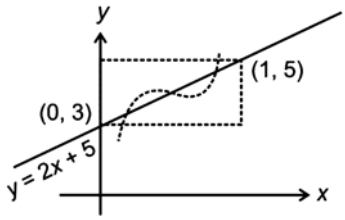
for minimum integral value $\alpha = -15$ and $\beta = 15$

$\Rightarrow \alpha^2 + \beta^2 = 450$.

Q.23 Let $f : [0, 1] \to \mathbf{R}$ be a twice differentiable function in (0, 1) such that f(0) = 3 and f(1) = 5. If the line y = 2x + 3 intersects the graph of f at only two distinct points in (0, 1) then the least number of points $x \in (0, 1)$ at which f "(x) = 0, is _____.

Ans. **[2]**

Sol.



If a graph cuts y = 2x + 5 in (0, 1) twice then its concavity changes twice

$\therefore$ f "(x) = 0 at atleast two points.

Q.24 If $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)}}}dx = \alpha\sqrt{2} + \beta\sqrt{3}$, where α, β are integers, then $\alpha + \beta$ is equal to

Ans. **[10]**

Sol. Put $x = \tan\theta \Rightarrow dx = \sec^2\theta d\theta$

$$\Rightarrow \ell = \int_0^{\frac{\pi}{3}} \frac{15\tan^3\theta \cdot \sec^2\theta d\theta}{\sqrt{1+\tan^2\theta+\sqrt{\sec^6\theta}}}$$

$$\Rightarrow \ell = \int_0^{\frac{\pi}{3}} \frac{15\tan^2\theta\cdot\sec^2\theta d\theta}{\sec\theta\sqrt{1+\sec\theta}}$$

$$\ell = \int_0^{\frac{\pi}{3}} \frac{15(\sec^2\theta-1)\sec\theta\tan\theta d\theta}{(\sqrt{1+\sec\theta})}$$

Now put $1 + \sec\theta = t^2$

$\Rightarrow \sec\theta \tan\theta\, d\theta = 2tdt$

$$\Rightarrow \ell = \int_{\sqrt{2}}^{\sqrt{3}} \frac{15((t^2-1)^2-1)\ 2tdt}{t}$$

$$\Rightarrow \ell = 30\int_{\sqrt{2}}^{\sqrt{30}} (t^4-2t^2+1-1)\,dt$$

$$\Rightarrow \ell = 30\int_{\sqrt{2}}^{\sqrt{3}} (t^4-2t^2)dt$$

$$\Rightarrow \ell = 30\left(\frac{t^5}{5}-\frac{2t^3}{3}\right)_{\sqrt{2}}^{\sqrt{3}}$$

$$= 30\left[\left(\frac{9}{5}\sqrt{3}-2\sqrt{3}\right)-\left(\frac{4\sqrt{2}}{5}-\frac{4\sqrt{2}}{3}\right)\right]$$

$= 16\sqrt{2}-6\sqrt{3}$

$\therefore \alpha = 16$ and $\beta = -6$

$\alpha + \beta = 10$

Q.25 Let $A=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}$, $\alpha, \beta \in \mathbf{R}$. Let α_1 be the value of α which satisfies $(A + B)^2 = A^2+\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$ and α_2 be the value of α which satisfies $(A + B)^2 = B^2$. Then $|\alpha_1 - \alpha_2|$ is equal to _____.

Ans. **[2]**

Sol. $(A + B)^2 = A^2 + B^2 + AB + BA$

$$= A^2+\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$$

$$\therefore B^2 + AB + BA = \begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix} \quad \ldots. (1)$$

$$AB=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}=\begin{bmatrix}\beta-1 & 1\\ \alpha+2\beta & 2\end{bmatrix}$$

$$BA=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}=\begin{bmatrix}\beta+2 & \alpha-\beta\\ 1 & -1\end{bmatrix}$$

$$B^2=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}=\begin{bmatrix}\beta^2+1 & \beta\\ \beta & 1\end{bmatrix}$$

By (1) we get

$$\begin{bmatrix} \beta^2+2\beta+2 & \alpha+1 \\ \alpha+3\beta+1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$\therefore \alpha = 1\ \beta = 0 \Rightarrow \alpha_1 = 1$

Similarly if $A^2 + AB + BA = 0$ then

$$\left(A^2 = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix} = \begin{bmatrix} -1 & -1-\alpha \\ 2+2\alpha & \alpha^2-2 \end{bmatrix} \right)$$

$$\begin{bmatrix} 2\beta & \alpha-\beta+1-1-\alpha \\ \alpha+2\beta+1+2+2\alpha & \alpha^2-2+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\Rightarrow \beta = 0$ and $\alpha = -1 \Rightarrow \alpha_2 = -1$

$\therefore |\alpha_1 + \alpha_2| = |2| = 2$

Q.26 For p, q, $\in$ **R**, consider the real valued function $f(x) = (x - p)^2 - q$, $x \in$ **R** and $q > 0$, Let a_1, a_2, a_3 and a_4 be in an arithmetic progression with mean p and positive common difference. If $|f(a_i)| = 500$ for all i = 1, 2, 3, 4, then the absolute difference between the roots of $f(x) = 0$ is

Ans. **[50]**

Sol. $\because$ a_1, a_2, a_3, a_4 are in A.P. and its mean is p.

$\therefore$ $a_1 = p - 3d$, $a_2 = p - d$, $a_3 = p + d$ and $a_4 = p + 3d$

Where $d > 0$

$\because$ $|f(ai) = 500|$

$\Rightarrow |9d^2 - q| = 500$ and $|d^2 - q| = 500$ .... (i)

either $9d^2 - q = d^2 - q$

$\Rightarrow d = 0$ not acceptable

$\therefore 9d^2 - q = q - d^2$

$\therefore 5d^2 - q = 0$ ... (ii)

Roots of $f(x) = 0$ are $p + \sqrt{q}$ and $p - \sqrt{q}$

$\therefore$ absolute difference between roots = $|2\sqrt{q}| = 50$

Q.27 For the hyperbola $H : x^2 - y^2 = 1$ and the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$, let the

(1) eccentricity of E be reciprocal of the eccentricity of H, and

(2) the line $y = \sqrt{\frac{5}{2}}x + K$ be a common tangent of E and H.

Then $4(a^2 + b^2)$ is equal to _____ .

Ans. **[03.00]**

Sol. The equation of tangent to hyperbola $x^2 - y^2 = 1$

within slope m is equal to $y = mx \pm \sqrt{m^2 - 1}$ .... (i)

And for same slope m, equation of tangent to ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $y = mx \pm \sqrt{a^2m^2 + b^2}$ .... (ii)

$\because$ Equation (i) and (ii) are identical

$\therefore a^2m^2 + b^2 = m^2 - 1$

$\therefore m^2 = \frac{1+b^2}{1-a^2}$

But equation of common tangent is $y = \sqrt{\frac{5}{2}}x + k$

$\therefore\ m = \sqrt{\frac{5}{2}} \Rightarrow \frac{5}{2} = \frac{1+b^2}{1-a^2}$

$\therefore\ 5a^2 + 2b^2 = 3$ .... (i)

eccentricity of ellipse = $\frac{1}{\sqrt{2}}$

$\therefore\ 1 - \frac{b^2}{a^2} = \frac{1}{2}$

$\Rightarrow a^2 = 2b^2$

From equation (i) and (ii) : $a^2 = \frac{1}{2}$, $b^2 = \frac{1}{4}$

$\therefore\ 4(a^2 + b^2) = 3$

Q.28 Let $x_1, x_2, x_3, ... x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is construction replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is _____.

Ans. **[142]**

Sol. $x_1, x_2, x_3, ..., x_{20}$ are in G.P.

$x_1 = 3$, $r = \frac{1}{2}$

$\bar{x} = \frac{\sum x_i^2 - 2x_i i + i^2}{20}$

$= \frac{1}{20}\left[12\left(1 - \frac{1}{2^{40}}\right) - 6\left(4 - \frac{11}{2^{18}}\right) + 70 \times 41\right]$

$\begin{cases} S = 1 + 2\cdot\frac{1}{2} + 3\cdot\frac{1}{2^2} + ... \\ \frac{S}{2} = \frac{1}{2} + \frac{2}{2^2} + ... \end{cases}$

$\frac{S}{2} = 2\left(1 - \frac{1}{2^{20}}\right) - \frac{20}{2^{20}} = 4 - \frac{11}{2^{18}}\Bigg\}$

$\therefore\ [\bar{x}] = \left[\frac{2858}{20} - \left(\frac{12}{240} - \frac{66}{2^{18}}\right)\cdot\frac{1}{20}\right]$

$= 142$

Q.29 $\lim\limits_{x\to 0}\left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x)}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)}\right)^{\frac{100}{x}}$ is equal to _____ .

Ans. **[01]**

Sol. Let $x + 2\cos x = a$

$x + 2 = b$

as $x \to 0$, $a \to 2$ and $b \to 2$

$$\lim_{x\to 0}\left(\frac{a^3+2a^2+3\sin a}{b^3+2b^2+3\sin b}\right)^{\frac{100}{x}}$$

$$= e^{\lim_{x\to 0}\frac{100}{x}\cdot\frac{(a^3-b^3)+2(a^2-b^2)+3(\sin a-\sin b)}{b^3+2b^2+3\sin b}}$$

$$\because \lim_{x\to 0}\frac{a-b}{x}=\lim_{x\to 0}\frac{2(\cos x-1)}{x}=0$$

$$= e^0$$

$$= 1$$

Q.30 The sum of all real value of x for which $\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2+5x+12}$ is equal to _____ .

Ans. **[06]**

Sol. $\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2+5x+12}$

$\Rightarrow \frac{3x^2-9x+17}{5x^2-7x+19}=\frac{x^2+3x+10}{3x^2+5x+12}$

$\frac{-2x^2-2x-2}{5x^2-7x+19}=\frac{-2x^2-2x-2}{3x^2+5x+12}$

or

$\Rightarrow$ Either $x^2+x+1=0$, No real roots

$5x^2-7x+19 = 3x^2+5x+12$

$2x^2-12x+7=0$

sum of roots = 6